

Vishnu Priya Nuthanapati

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Summary

Computational bioinformatician with a master's in bioinformatics, focused on cancer genomics, tumor immunology, and multi-omics integration. Experienced in developing and applying reproducible pipelines for bulk and single-cell RNA-seq, spatial transcriptomics, epigenomic profiling, and genomic data. Performs quality control, statistical modeling, machine learning, and integrative analysis to surface biologically meaningful patterns and generate testable hypotheses. Skilled in unsupervised clustering, differential expression, data visualization, and the inference of cell state, gene regulatory networks, and cell-cell interactions. Programs primarily in R, Python, and bash, working closely with experimental researchers on study design, interpretation, and manuscript preparation.

Relevant Technical Skills

- R/BioConductor/RShiny
- Docker/Singularity
- Snowflake
- JupyterHub/Notebooks
- SQL/MySQL
- Cloud Services (AWS)
- HPC Computing (Slurm)
- Mage/NextFlow/wdl
- Web Technologies
- Python/BioPython
- Conda/PositConnect
- Rest APIs
- Unix/Shell/Bash
- Pandas/Streamlit/Plotly
- HTML/CSS/NodeJS
- GitHub/BitBucket
- Matlab
- Rshiny

Domains Worked In

- NGS/RNA-Seq (small, bulk, sc)
- Transcriptomics
- Mutation Analysis (Vartrix)
- Benchling
- CITE-Seq
- WGCNA
- Probability & Statistics
- EHR (AllofUs)
- Spatial (Geomx/ VisiumHD)
- Machine/Deep Learning
- Visualization (Cytoscape)
- Pacbio/NanoPore

Work Experience

University of Mississippi Medical Center

Jun 2025 – present

Bioinformatician 1

Mississippi, USA

- Analyzed ROI-based spatial-transcriptomics data with CellChat to map region-specific expression and cell-cell signaling, identifying 30+ ligand-receptor pairs and spatial niches to prioritize pathways for targeted validation.
- Built an interactive R Shiny CellAtlas for placental marker exploration, enabling real-time visualization of 20,000+ single-cell profiles and reducing marker query and export time by 40%.
- Integrated DNA-methylation, variant, and ClinVar annotations to prioritize clinically relevant loci, flagging 150+ candidate sites and accelerating patient-centered genomic interpretation by 35%.
- Quantified differential exon usage across conditions to identify alternative-splicing changes, uncovering phenotype-associated exon patterns in 10+ key transcripts.
- Designed reproducible, well-documented small and bulk RNA-seq differential-expression pipelines with mixed-effect models, enabling robust expression and pathway analysis across 200+ samples.
- Assembled de novo transcriptomes for non-model organisms with limited genomic references, strengthening downstream analysis.

Vor Biopharma Inc

Jan 2024 – Jul 2024

Quantitative Biology Co-op

Massachusetts, USA

- Built Nextflow pipelines to analyze long-read NGS data and characterize AML mutations at single-cell resolution, improving detection sensitivity by 15%.
- Traced AML clonal evolution from diagnosis to relapse, identifying 20 additional mutations that clarified disease progression and emerging therapeutic resistance.
- Built a data-validation pipeline that enriched sequencing data and reduced error rates fivefold, improving the reliability of downstream genomic analysis.
- Built ETL pipelines in Mage that cut data-processing time from 12 to 2 hours and served thousands of records daily through REST APIs.
- Designed and deployed sample-inventory dashboards with the process-development team, raising process efficiency by 40% through real-time tracking.
- Evaluated CRISPR adenine base-editing (ABE) edits from NGS data, building visualizations that let wet-lab scientists review 200+ samples daily.

- Built a registered sequence-listings database with the business-development team, streamlining access to 5,000+ records and improving IP-management efficiency by 30%.

Accenture Private Solutions

Aug 2021 - Jun 2022

Associate Software Engineer

Telangana, India

- Automated DocuSign access requests with Power BI alongside the DevOps team, processing 50+ requests daily.
- Led server patching for client operations, covering scheduling, testing, and deployment of updates across client systems.
- Managed Global Asset Management System (GAMS) user accounts in SQL, improving request response time by 25% and account-management efficiency by 30%.
- Administered global asset permissions, controlling access to terminated-employee records and asset tracking.

Campus Fellowships Mentorship Program

Nov 2020 - Jun 2021

Bioinformatics Research Intern

Telangana, India

- Processed NGS data with RSEQREP on AWS, optimizing pipelines to handle thousands of samples and improve analysis accuracy.
- Optimized transcriptomic pipelines spanning QC (FastQC, MultiQC), alignment, and expression quantification for high-quality large-scale downstream analysis.
- Conducted downstream genomic analysis, identifying key regulatory pathways and novel gene targets through differential expression and pathway enrichment.
- Built the mentorship-program website with HTML, CSS, Node.js, and Balsamiq, driving a 35% increase in student outreach.

Knowledge Solutions India

Aug 2020 - Oct 2020

Research Intern

Hyderabad, India

- Built an insurance-premium prediction engine using an ensemble of multiple linear and random-forest regressors, improving prediction accuracy to support decision-making.
- Applied PCA-based feature selection, achieving 1.5% MAPE on premium predictions.

Gowri Gopal Hospital

Aug 2020 - Oct 2020

BioMedical Intern

Kurnool, India

- Launched an orientation program on basic medical-device error handling for interns and doctors across all departments.
- Calibrated medical devices and instruments with standard calibration kits, reducing the error rate by 5%.

Projects

Mapping of miRNA-Gene regulatory Networks in Schizophrenia Across Tissues

Jul 2021

- Built a miRNA-gene regulatory network from TargetScan 7.1 predictions across 5 brain and blood tissues, resolving a 2,813-gene, 50-miRNA network in the BA46 prefrontal cortex with Cytoscape.
- Mapped 7 dysregulated hub genes to 4 repurposable drugs (Valproic Acid, Aspirin, Estradiol, Vorinostat), prioritizing schizophrenia drug targets with minimal cross-tissue side effects.

Single-Cell Transcriptomic Profiling of Ribosomal Dysregulation in Tumor-Associated B-Cells

Aug 2024

- Resolved 12 immune cell populations from 13,962 murine lung cells by building a reproducible R/Seurat pipeline (QC, doublet scoring, Harmony batch correction), surfacing tumor-associated neutrophil and macrophage states.
- Identified 1,734 differentially expressed genes (FDR less than 0.05) in tumor vs. healthy tissue via Wilcoxon testing, linking them to neutrophil-activation programs through GO/Reactome enrichment and GSEA.
- Mapped 523 ligand-receptor interactions across 28 signaling pathways with CellChat to characterize the tumor immune microenvironment.

Education

Northeastern University

Boston, MA

Master of Science in Bioinformatics; **GPA: 3.9/4.0**

Dec 2024

- **Coursework:** Bioinformatics Computational & Statistical Methods, Advanced Genomics, Omics, ML & AI

Osmania University, College of Engineering

Telangana, India

Bachelor of Engineering in Bio-Medical Engineering; **GPA: 4.0/4.0**

Aug 2017 - Jul 2021

- **Coursework:** Biomedical Instrumentation, Cell-Tissue Engineering, Human Physiology, Biomedical Signal Analysis

Certifications & Conference & Volunteer Activities

- Obtained AWS Certified Cloud Practitioner certification
- Presented Poster at the Experimental Biology 2021 Conference
- Volunteered at IEEE, Hyderabad section, as a marketing volunteer (EMBS society)